

# ActInf Livestream #036.2 ~ "Modelling ourselves: what the free energy prin

*Livestream | 1:51:34*

Hello and welcome everyone.

This is ActInfLab livestream number 36.2.

It's January 26th, 2022.

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We are a participatory online lab that is communicating, learning, and practicing applied active inference.

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This is a recorded and archived livestream, so please provide us with feedback so we can improve our work.

All backgrounds and perspectives are welcome and we'll be following good video etiquette for livestreams.

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Today in 36.2, our third discussion on 36, we're going to continue to learn and discuss this paper by Sims and Pizzullo 2021, Modeling Ourselves.

What the Free Energy Principle reveals about our implicit notions of representation.

And we're going to have a nice jumping off .2 as we do and have some questions written down and prepared.

Other things will spontaneously arise and if anyone watching live wants to just write a question in a live chat, we can also address that.

We'll start with an introduction.

So we'll go around and introduce ourselves.

People can say anything they want and they can also add just what got them excited about the paper in general or for this .2 discussion specifically.

So I'm Daniel, I'm a researcher in California and just excited to see how this representation discussion influences our own self representation and that at the individual and the lab scale.

Moving forward, I'll pass it to Blue.

Hi, I'm Blue.

I'm an independent research consultant in New Mexico and I am excited by this paper because in our previous discussions we're as usual left with more questions than answers.

And also I kind of am finding myself like swayed by the different like representational, structural, content driven, non-representational, like these different arguments and I just find them very interesting and I just want to know if at the end of this, if I'll be swayed more or less in certain directions or maybe just be more open to other interpretations of FUP.

And I will pass it to Danielle.

Hello, I'm Danielle.

I'm a cognitive scientist at Google with a background in language evolution, language development.

I'm really interested in thinking about how humans evolved the ability to model other minds.

So interested in how this discussion can contribute to that and generally interested in how really good frameworks like the FUP can help kind of encompass other frameworks and revise them.

Awesome.

Dean?

Hi, I'm Dean.

I'm up here in Calgary in Canada and a little bit of recency effect.

I started watching Ozark the last season and so I am trying to plug this paper in through what would Marty Bird do?

So that's kind of my little twist on modeling.

So back to you, Daniel.

Who is Marty Bird and what would he do?

Right.

Exactly.

Great question.

Well, that's all we prepared for today.

So we'll go into the discussion.

Okay.

But we have some things written down.

Is there anything just off the bat that anybody wants to just ask or jump into or like a figure or a quotation to begin with, or we can look at some of the things that we had written down.

Okay.

Let's just start with what we've written down.

And then of course, at any point we can branch off.

Okay.

Okay.

I think a big theme that we will probably return to again and again is this idea of moving back and forth from non-representational forms to representational forms.

In the paper, they introduce four different facets of representation and then how each of those facets can be approached in a representational or non-representational way.

And although the image here shows some solid black lines, that was kind of the discussion was, is there movement across these areas or blurriness?

So maybe, Dean, what is there to say or start to explore with representations emerging, like almost precipitating or crystallizing out of something non-representational, and then the reverse process of something that is more representational submerging back into something that's less?

Well, one of the things the authors did talk quite a bit about in terms of the functional aspect of this was the vicarious nature.

And then they also talked quite a bit about whether or not somebody who's sort of moving in that radical inactivism space still has to be able to differentiate and get some, include some of the terror temporal aspects of what might pop up and become something that we can stabilize and in model representation form.

But then it has to also be able to disappear, right?

It also has to not necessarily consume everything in terms of our attentional fields.

So I'm not really sure exactly how that works itself through.

But I would think that it's difficult to say that even if we're in a flow state, that there isn't some moment when we do reflect in that mirror.

So that's kind of where I'm at.

I don't have any answers per se, but I don't want to get stuck in the idea that it's just an on-off switch.

I think that there's a bit of a moment of dimming and then re-lighting.

So.

That makes me think about certain paths that our thought takes over these eight cells.

Like when we're in the representational side, when we're dealing with the representation, how do we take that tetrahedra and look at the four different sides?

And then how do we stay within a column looking at just the organizational aspects of a system and then move north and south on that?

Do these kinds of knowledge or cognitive transitions happen all the time?

Do they have other names that might be more familiar than using some of this philosophical language?

Like what is it?

Are there any times in our days where we're dealing with something representationally and we shift to a non-representational version within one of these categories or across categories?

So I was just thinking of that actually, like, what does an organizational representation look like?

Like that is like a map, like a structural representation, a content related representation, and a functional representation.

And like what do these representations look like?

Like, and how are they different?

I was just thinking about that.

Let's go to the definition of the organizational representation just to kind of remember.

The fact that FEP requires internal states, states that encode the recognition model that are statistically separated from the external reality.

So the organizational area is about how variables inside of systems are separated from variables outside of the system.

So the pro-representation take is there's an evidentiary boundary, the Markov blanket, and there's something on the inside that is doing something like a representation on what's outside.

So that's the pro-representational take.

Then the non-representational take is where we get into the ecological and the inactivist persuasion.

And that's where the authors say that some have approached this Markov blanket structural formalism, even though we're talking about it in the organizational, not the structural facet, in a manner that suggests a non-representational view.

And they argue that just partitioning two coupled systems like agent and environment with a Markov blanket doesn't imply that the behavior of the agent is explained by anything related to an internal model that's doing anything like capturing the structure of the outside world.

So you just use like organizational and structural like in this like big hobnob.

But so my question was really, what does an organizational representation look like?

And so I'm hearing that you answered that there are clearly defined boundaries in organizational representation or partitions between internal and external states.

So like I have a clearly when I have a representation in my mind that's organizational, it is like your boundary is different from like, is it only your own boundary?

Or is it like the boundaries of each individual agent are organizationally represented in your mind?

Or is it just the boundary between you and the outside world?

And how might that differ from structural and content related?

Like what do these different representations look like?

That's what I'm trying to see.

Like, would they be maps and how would the maps be different?

The structural, which does come up several times in the organizational definition.

So we see structure coming up, the structure of the system in how it's organized.

But what is the structural specifically?

Okay.

Having representational vehicles that are structurally similar to the state of affairs in the world that they stand for.

The pro representational structural side is kind of like the good regulator theorem of cybernetics.

So the organizational side is highlighting that there is some sort of informational encapsulation, we might even say, like our previous discussions, but some sort of organization of variables such that there's an inside and an outside that are separated.

And the structural side is saying that it's like if there's three things that are connected in the outside world, then not just that there are variables that are organizationally separated in the internal states, but that the structure of those isolated variables is going to do something like recapitulate or have a structural resemblance to the generative process.

So that's the generative process.

Generative process, that's like the niche, that's what's outside generative model, that's what the agent has on board.

And so if it's to be useful for control purposes, then it may have to have some sort of structural resemblance to the actual connectivity of the generative process.

process. So generative model and generative process being separated is what's captured in the organizational side. Them having similar resemblances or similar structures is what the structural side is. The structural non-representationalists are suggesting, this is I think Tale of Two Densities with Ramsted et al., or modeling ourself.

The authors suggest that generative models do not meet the requirements of structural representations because the process of exploitable structural representations, so posterior beliefs in the entity, is enacted. And so it doesn't necessarily have to recapitulate the form of the generative process. So maybe one could be skilled in the performance of driving without having variables connected in the way that a car pieces are connected.

It comes back to a theme that Active presents to us, perhaps more as a question than an answer, which is how much do we have to know in order to act? Do we have to have an internal model that is very similar to the process? Uncorrelated in how similar it is? Anti-correlated in how similar it is? And can we have a framework as researchers that helps us compare the structure and organization of generative processes and generative models of those processes?

Yes. Go for it, Dean.

So I'll put this out there and then people push back or say it doesn't make any sense.

From what I read in the paper now, because I had to go back over and look at it again, if I were to try to draw a simple comparison between organization and structure, organization would, you could see it as being something as being either in or out. So you could be inside, you could be inside a cell or outside of a cell, then in our minds, we can tell, we will draw those partitions and then we'll decide what's in or out. Structural, I think, takes on more of the entailing question, something that Sir Ball brought up in the chat, which is, so once you've decided whether something's in or not, is there something else that now can be seen as being superordinate to that from us from, from the perspective of sort of what's wrapped around now the thing that's in, or what can sit upon theoretically the thing that you've now decided is foundational to whatever it is you're looking at and what can be set aside as not being important to what you're looking at. So again, pull that apart and say that that's wrong, but that's kind of what I try to read into what the readers or what the writers were saying. So it's just my interpretation.

Thanks, Blue.

So I just wonder in this organizational representation, do you draw a distinction between, I mean, clearly there's one between like what's internal and external to yourself, but are you also trying to like draw distinctions between what's internal and external to your computer or your refrigerator or your best friend or your child? Like, are you making a representation of everything that's internal and external to every other thing? Or is it only with respect to your own self versus environment? I would think it's when you do that. I don't know that it's implied that you do it all the time. But if you choose to do that, I think that that's something that's pretty easy to tell the difference

around. So again, I'm not sure that they were saying that we do this, we're always constantly doing this, but I think what they were trying to do is show the difference between knowing when something's in or out, versus knowing when something's first or second, from a structural standpoint. Again, I don't want to put words in the author's mouth. But I did hear that in the words that they were doing to try to parse those two things and give each their due, as opposed to saying they're just the same thing overlapping. Let's try one little thought loop. So let's start with Blue's question of what does an organizational representation look like? So we'll start in that very top left cell, and then shift the focus towards the structural components, and then challenge the structural representation, but stay within the structural domain, and then see if that can result to us making it back to a non-representational organizational framing. So what is a good organizational? What's something that is classic vanilla organizational representation? Well, that's something that is well separated in terms of variables inside and outside the system. So we're not worried about the structure of the variables inside or outside, just that they are separated. So computers seem like pretty clear cases where within a program, you could have variables that by design are separated through like an intermediate variable or computers and hardware. Like two computer systems could be totally separated, except for an interface like a USB port or something. But of course, this paper is about that strange loop when it's a cognizing agent, an adaptive active inference agent doing that representing, not necessarily a mere active agent, but should we use a computational example or some sort of enacted human example? You got a good example right here on the page. So how would we shift across the y-axis between organizational representational, because I'm assuming we're going clockwise, to representational structural? That's not a hard example to make.

Yes. So in the human case, this would be like two humans in a conversation.

Mm-hmm. They are in a conversation. Yes. Not outside of the conversation.

Outside of the conversation. Yes. Two humans, when two humans are in conversation, there is separation in terms of their sensory motor systems. So we have checked the box for organizational representation, starting in the top left cell. All right. Now the question would be, is there structural representation? So that would be like, maybe if one person thought of a sentence and then said it, and then the other person thought of that sentence too, would that qualify as a structural representation? Because there's a structural resemblance in the model of the two convergence? Yeah. We could also find out if both of us were talking at the same time and suddenly stopped turn-taking, that the structure of that model would now fundamentally change. It's still representational, and two people arguing over top of one another to the third party isn't necessarily something comprehensible. Never mind to the two people that are inside the conversation, right?

Blue? So when I think about conversation, something comes to mind recently. Like really, you need to have the order of the language specified. Like my son is reading now, and he's five, and he's taken the book like pop on pop, the Dr. Seuss book, and he'll turn it upside down, and he'll say, Dodd no doe, and like, he'll read it that way. And then he wants to read like every sentence backwards, like from the end to the beginning, which I'm like, I don't care, as long as you're reading,

because it gets him practicing reading. But it's completely, the context is completely different, right? When you're reading like each word in the opposite order. So like when you're talking about a structural representation of a conversation, I think that there needs to be like a literal representation, or at least like a coarse-grained representation of in what order things are said in order to glean some kind of meaning. But then are we bleeding into the content related? But like, as we're, you know, that's why I really feel like, like, what are these different maps look like? And trying to maybe elucidate what the representations are is valuable. Because I think when you have organization, okay, inside, outside, like this is the sentence, or he said this, and she said that, or whatever. And then you try to put structure on it, like, that gives it then content or context that they can then, you know, and so on and so forth.

Dean?

No, I don't have much to add to that other than, I just don't think that the, that this loop that we've, we've just selected this, we said there was going to be certain amount of randomness in the last live stream. And we've chosen this now, let's walk, let's walk through it. I don't see it being difficult to follow that path and find examples. So we have a superordinate right now, structurally, we want to carry out this directionality from the representational side of things to the non representational side. And, and blue, if you're if your kid wants to turn the book upside down, that's, that's actually a feature, right? Because they're not already stuck in this only one way of doing it thing. So, yeah.

Dean?

Dean?

Let's complete the red loop. And then, then the green and the orange will come into play. Okay. So we started with two people in conversation. That's organizational separation of cognition.

Dean? This brings us to the structural facet, which is where there's a structural resemblance in the representations of the conversants. So there, maybe it is the case where they're both thinking about a similar topology of variables. So there's some structural representation, but now let's stay within the structural column, go from B to C. Okay. So the two people are in a conversation. It's kind of like a dance. So what if those two people's representation is so different that it's actually not structurally resembling? So maybe one person is the more experienced in the discipline and the other is less experienced. Like one person's super good at fixing motors. The other person is just has no idea. So the representationalist take would be like, well, they're both thinking about the motor, but then when we start to see like a cognitive asymmetry, all of a sudden we fall out of structural representationalist between these two systems because their cognitive models don't necessarily have similar structure.

So I might be missing something here, but I thought that we can only start to talk about the structural dimension here once in this example, the two people in the conversation are representing the same thing. And then the question is how much information or how does each of their representations do each of their

representations have to look similar to one another? But like part of it has to be that they are representing the same thing. Is that a sort of necessary component? If they're not representing the same thing, then we're having a different conversation.

Dan Heldon Well, that may come to the functional

that might come to the content related like the novice and the expert. The representations are both about the engine, but there's so structurally different in their representation of the engine so that although the content in the aboutness of the representation is similar. So still qualifying as a representation in that sense, there's a structural non resemblance such that by that criteria in that situation, we've fallen out of the representational cell B into something non representational potentially blue. What do you think?

So I think like two people in a conversation can be trying to represent the same thing like like, yes, they both need to be

trying to represent that have the same representation like they're trying to converge on a conversation, a meaningful

conversation, presumably. But they don't always converge. And then that's where you get like misunderstandings or or like, you know, we both think that we think the same. We both agree that we're thinking that we have the same structural representation,

but perhaps they're different and then and then the communication breaks down.

Okay, so the structures are so different that we're recognizing it as a continuum from total overlap of structural resemblance to total non overlap, but it would be important to specify that null hypothesis. So we've kind of scooted into C.

C. How do we stay in that space of like the novice and the expert who have different cognitive representations and then use that to challenge even organizational

representational

representationalism?

Can I

Blue and then

Oh, Dean first and then

Go ahead, Dean.

Right. I just thought I want to just go back to the to Daniel's point. So if we're if we're structuring a conversation normally,

we turn take, but if we're playing in a band,

we're all playing at once and then we stop playing. So I think we can start with structure to is just deciding how we want to signal.

Right. So I don't think it I don't think there's a necessarily an order that we have to start with organization first, we can actually start with structure first as well.

Yeah, and I think a conversation is a particularly challenging example because we're representing two things what we're representing with the other person is trying to stay and then presumably what the conversation is about is something else in the world that we could be representing.

Exactly. Yeah.

Blue.

So I think we're representing a degree of information sharing and I think that this is going to like lump into autopoiesis, which I've been really wanting to kind of get into. But but when we have like I'm my own person and I'm communicating with Daniel from you know, 50,000 miles away, we both feel like we're getting on the same page when we actually like do get on the same page.

Like do we form some kind of like separate cognitive unit like at what degree of model overlap is the degree of information sharing so high that then we are worthy of our own Markov blanket like we're sure that we understand that we are the same unit.

Right. And I think that this idea of self assembly and information sharing and model overlap, like I think that there's an important path to traverse down this way because if you don't have full overlap of the model like you're or do you need full overlap to form a self assembling system to be, you know, to form a higher level. It's just an interesting thought.

That's kind of where the C to D transition takes us, which would be like, so we have the drummers waiting for this and has a model resembling this and then the singer has a totally different structure.

And then when we think about the whole cognitive system, it doesn't deny what we recognize in a that there's still is like the sensory motor separation.

But we've reached that point by those people or roles being part of a larger non separable extended cognitive system.

So almost by recognizing the interaction.

So here we focused on the separation in a and that's what granted us the representational cell, but we've returned to seeing that partial information encapsulation within some type of broader structure where none of the entities have like the band level representation.

So hopefully other people can think of other like ABCD examples.

But when we were at B, we kind of started branching off into like talking about the content related things, the aboutness.

So here it's like two people do have a structural resemblance in their internal model and it's about the same thing.

But then what if one person sees the aboutness in such a different way that again, we fall out of the content related representation.

And then how would we move from there back to C where all of a sudden the cognitive models in a sense don't have structural representations, resemblance because they're about different things now.

Danielle.

So would this be like in the case of two, two things that are so analogous that they share a lot of structural similarities and things like Dedry Gintner's work that you can actually draw lines between the individual aspects of the concepts, but they could be about totally different things.

But two people talking about these things can form some sort of structural similarity in the representations and it's just like an entry point into being able to think about all other things.

But the aboutness really has nothing to do with it.

But the aboutness really has nothing to do with it.

Hmm.

Dean, then I might want to ask me on that.

Dean, then I might want to ask me on that.

Dean, then I don't know.

I don't know if this answers that, Danielle, but in my space, I'm thinking the difference between say a distracted driver who's looking down at their phone between the seats and somebody who's got that information and heads up display on their windshield.

Right, so so content twice but structurally presented one as a one as a distracted state and one as a contiguous state.

So I don't know if that answers your point, but I think we could, we know the difference right in terms of what where the where the content is structured and how we're trying to contextualize each each matter.

So.

So.

So.

Again, staying with this sort of educational setting.

What is it important to see a transference or a new emergence of?

Is it that the structural model of the learner is moving into more resemblance with the teacher?

Is it that the aboutness of the learner is moving towards the same aboutness of the teacher?

Or let's just remind about what the functional is.

Supporting vicarious use before or in the absence of external events.

So does the teaching conversation, will it be a functional representation when vicariously, without the teacher, the student can carry out what the teacher could do?

Which might be having aboutness or not.

It might be with a structural cognitive resemblance or not.

It might involve total informational isolation, putting a squarely in A, or it might involve some type of challenging of that in D.

The reason why there's not going to be, I think, a precise answer, because as the paper lays out, even for the same scenario, people do have different perspectives.

So it's not like we're trying to take a scenario and classify it into, well, that one's like an A, B, E, G, or an A, C, E, H.

Like, that is probably not what one of the outcomes could be.

But these are all perspectives that we could take on a given scenario.

And they do reveal interesting things about the systems of studies.

Anything else to add on this sort of, like, looping around the cells?

Or should we push on?

Push.

Push.

Okay, Blue, what were you thinking here?

I was just taking notes.

And so, like, if, so maybe we should talk about content related and functional.

So I was just taking notes on, you know, what these are organizational, like, how are the representations?

And I was trying to draw a picture.

So if you, I will draw my picture if, if you will talk about maybe functional representation, like, what does a functional representation look like?

Or content related?

What is it, what's inside of those?

And then I will draw my picture and then you can show it.

Or I'll just draw it while you're talking.

Sure.

So the content related is whether the generative models need to explicitly model the ways external states produce sensations, environmental models, or the ways actions produce sensations.

Is it really important that internal states resemble external states?

Or is it sufficient that they afford accurate action control?

So, let's just say there's a spotlight that's shining photons onto the eye.

The content of the cognitive representation would put us more, we'd be more in the representational camp if the cognitive model were truly explicitly modeling the ways in which the photons hit the retina.

Maybe.

Maybe.

But this or is a bit challenging because the ways actions produce sensations, that could be a model of like, well, if I blink, it becomes darker.

And when I open my eyes, it's lighter.

But that's a totally disjoint question from understanding how a spotlight works.

But those two scenarios of the spotlight's cognitive explicit modeling, the environmental model, or the sensory motor model of blinking and it becoming darker, those are both content related.

And Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

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Dean?

Dean?

Dean?

Dean?

Dean?

there, but let's go back to just review functional. Like what is the functional? And it's good to revisit these in the zero, one, two, three, four, five, because it's the contribution of the paper.

And maybe there are other facets, other columns to add. And there's just a lot of complexity, even as this is. So the functional one

is about supporting vicarious use before or in the absence of external events. So in that story with the lights and the blinking, that model only has to be instantiated while in that room with the photons hitting the eye and the blinking happening. However,

we can imagine a situation that's vicariously detached or before that setting. Like what would happen if there were a light over here and a light over here, and it was the case that blinking made it darker. And this is where they trace back to Piaget that representations should vicariously stand for something external in their absence and afford vicarious operations.

So there's a light over on the left side, you reach over and you unscrew it.

What is in your hands?

Now that is

seeming to be in the functional

facet because we're talking about

the role the representation is playing

and the fact that it is standing for

something external in the absence. There's no physical light that we're talking about.

It's not like we're unscrewing it and then verifying that it's in our hands.

Who's for and against that?

What functional roles do internal models play during free energy minimization?

And does minimization require the internal manipulation of variables

variables in ways that resemble vicarious operations in the classical Piaget account of representation?

Yes, Blue.

So this might be like, I don't know, maybe it's the easiest and also like the hardest to get a grip on.

So I think about  
like do we  
so if we have a representation of internal variables  
that like are functional, like a functional representation of  
a functional internal representation  
is this like doing  
like you manipulate the variables internally  
and then manipulate them externally  
like you plan to reach for the cup  
to take a drink of water  
and then you reach for the cup to take a drink of water  
but you have to have like the internal execution  
prior to or  
maybe simultaneous with  
the external execution  
is that like  
functional representation?

Like you plan the action  
and do the action  
maybe separate  
or maybe together  
is that what functional representation looks like?

Let's go with a reaching for the cup  
because as Danielle noted  
like

conversation  
improvised  
spontaneous conversation  
of reflexive entities  
is one of the hardest  
cases

but  
it's the one that we have  
so that's awesome

but  
we're going to be talking about a  
adaptive active inference agent  
reaching for a mere active inference agent

so  
in the Sims  
other paper we discussed  
now we're talking about the case of  
unidirectional  
integration  
rather than  
the multi-scale  
reciprocal integration  
so  
the person is separated from the cup  
there's a separation  
of the cognitive model  
of the agent  
from the cup  
the structural side  
if the internal model  
is  
cup-like  
or it features  
a cup variable  
touching a table variable  
and then  
another edge  
that is engaged  
of the hand  
touching the cup  
for example  
that might be a structural similarity  
but  
I think that's part of the  
debate is  
how could there be a structural similarity  
when  
it's not a cup  
in your brain  
which we kind of talked about  
last time

like  
how could you have something  
that structurally resembles a car  
if  
it's your brain  
and body  
the content related  
encoding environmental contingencies  
or sensory motor contingencies  
so it could be the case  
that a certain sensory motor action  
will result in picking up  
and grasping the cup  
whereas another sensory motor action  
is going to result in the shattering  
of the cup  
and that could be analyzed  
in terms of  
correctness  
the functional  
representational question  
would be like  
the vicarious  
detachment  
of the sensory motor loop  
from the cognition  
so then it would be like  
if you were to pick up that cup  
then  
you dumped it  
what would happen  
and then someone says like  
well water would fall out  
so in that case  
it is like  
there's a functional representation  
whether or not  
it resembles

the structure of the cup  
there's a functional representation  
because  
that  
idea  
of grasping the cup  
it's able to be operated on  
internally  
vicariously  
in the absence of direct stimuli  
related to  
the cup  
Dean  
do you mind if I just read  
right from the paper  
for a second  
because I think it helps  
in terms of this  
difference between the  
sort of the  
automatic  
I don't know  
automaticity  
and reflective  
aspects of this action  
okay  
one important implication  
falling out  
falling out of this diagnosis  
is that when  
considering functional  
rural aspects  
it's often how the details  
of our chosen process theories  
are fleshed out  
and contextualized  
by the kinds of cognitive phenomena  
that we are attempting

to account for  
for that skew  
or that  
for that skew  
our interpretation  
of FEP  
in one direction  
or another  
for example  
one may consider  
that there are core aspects  
of FEP  
such as  
the possession  
of a Markov blanket  
and more ancillary  
aspects  
and I think ancillary  
is the key here  
such as the possibility  
but not the necessity  
to engage in counterfactual inference  
which is only required  
for planning  
and it is only the latter  
more ancillary aspects  
that call for  
a representational interpretation  
under a given process theory  
this would imply  
that when using functional role  
as a sole criterion  
for representational processes  
only some FEP agents  
namely those  
that can engage  
in counterfactual forms  
of inference

would meet the criteria  
for representation  
it is only this subset  
of FEP agents  
that would be equated  
to full-fledged  
predictive processing agents  
so there is an automaticity  
part to this  
where it's really  
not a Markov blanket  
you just do it  
almost subconsciously  
and that's  
again the authors  
are trying to point this out  
yes if we fail  
I drop the cup  
and now I reflect on that  
that's the representational piece  
but prior to that  
I was just  
going through the motions  
I didn't need a set  
of instructions  
telling me to reach out  
for the cup  
so that's just  
in the author's words  
thanks a lot  
for clarifying that  
it's like  
yes if one is  
reflexively  
grasping at it  
or accidentally  
then  
it

then grasping the cup  
like if you're  
if you're groping around  
in a dark room  
and then  
you happen  
to grasp  
the cup  
it doesn't  
have to be  
representational  
because  
there isn't  
a counterfactual  
so  
it  
and that's  
very interesting  
how like  
they say  
that it is  
ancillary  
so that's  
kind of like  
secondary  
or  
not  
essential  
to it  
because we can  
imagine an active  
inference entity  
that's  
taking  
in sensory  
observations  
updating its  
generative model

engaging in  
policy selection  
and then  
resulting in  
some  
action  
output  
impacting the  
niche  
and the cycle  
begins again  
that can be  
just like a  
single layer  
model  
that doesn't  
engage in  
counterfactuals  
but  
they  
raise that  
notion  
that it's  
only  
the  
counterfactuals  
in  
cognition  
representation  
that actually  
like give  
us the  
space  
to have a  
functional  
representation  
it's like if  
the shell

just represents  
the shell  
can't be  
anything else  
but then there's  
another kind of  
entity who can  
see the shell  
as financial  
or can see it  
in a different  
way and then  
it's those  
counterfactuals  
that enable  
the shell  
to play a  
functional  
representational  
role  
in that it  
stands for  
something  
else  
blue what  
is this  
imagery  
on the  
bottom  
so I  
just was  
like trying  
to really  
look at  
or like try  
to visualize  
what would  
um these

different like  
representations  
really look  
like right  
so if we  
take like  
the idea  
of two  
a person  
and a  
cup  
even right  
okay so  
on the  
first one  
right we  
have  
organizational  
so like  
I am a  
thirsty  
body  
I'm contained  
within me  
there is a  
cup that  
contains water  
over there  
like I know  
my boundary  
I know the  
boundary of the  
cup and the  
representation  
looks like  
this um  
and then a  
structural

representation  
um is just  
me and  
the cup  
and my  
relationship  
between me  
and the  
cup  
right like  
so do I  
have to  
also have  
like like  
so my  
real question  
here is  
do these  
perhaps build  
on one  
another  
so so here  
you have the  
structural  
and um  
then the  
next one  
would be  
content  
related  
do I  
also have  
to have  
this  
connecting  
line in  
the content  
related

uh is that  
required  
like do I  
have to have  
an idea  
of the  
structure  
to understand  
the content  
and then  
um in the  
functional  
like do  
you need  
to know  
the content  
and the  
structure  
and the  
organizational  
to have  
to have  
an understanding  
of the  
function  
right do  
I need  
to know  
like the  
content  
related  
like I  
know that  
there's  
water in  
the cup  
right I  
assume

that there's  
water in  
the cup  
to find  
out I  
have to  
realize  
that there's  
an inside  
of me  
and outside  
of me  
and inside  
of the cup  
and outside  
of the cup  
I have to  
realize my  
relationship  
between me  
and the  
cup  
and then to  
find out  
the content  
like I  
have to  
go sample  
the cup  
like to  
see what's  
inside of  
it and  
I have to  
know what's  
inside of  
me and  
drink it and there's like whiskey in it or something like i'm not going to understand

the function until i have all those things layered on top of each other and so that's kind of really my question um here and daniel's drawing on my drawings so but but um really like i just wonder if there's levels of increasing complexity between these different kinds of representations or if it's just merely the structure the organizational and i don't need these boundaries between self and cup in these other models maybe it's just the line without the circles in the structural or the no external circles and content related so i just want to know like what you guys maybe think of that if there is a layering here because i was kind of layering them up but i don't know if that's really the right mental representation thanks blue dean

um i'm not muted good if you i think that i think there's two parts to this one is that we are able to differentiate which is what the table allows for and i think that the second part is the the arrow the curved arrow that daniel drew which essentially represents boundary crossing number one so the difference between something that's static and something that is translatable and then the second part of it i think without going back down the rabbit hole that we almost fell into in the point one of this when we started talking about free will and can and will and the orthogonal piece of that i think what it what it speaks to maybe is that it's not a question of free or or encapsulated it's it's a degree of dependency question when you when you throw that curved line on on this sort of tiled representation and start moving across a boundary so i don't want to call it degrees of dependency will up to a point of passing through zero but i'd like to because i think there is an aspect of that and that's why i think it's kind of it's kind of crazy if we just focus on the static things instead of the moving around the moving through four squares is what i really think gives our minds a good workout yep so here's another little um workout so blue had two circles in relationship so here the circle is going to reflect an adaptive active inference agent or the full-fledged predictive processing agent one that can engage in counterfactuals so the top is two adaptive entities in relationship and the triangle is going to reflect a mere active inference entity like a cup or just something that doesn't engage in counterfactuals so organizational is the separation of the systems so that's very similar because it's just describing there being a markov blanket separating the systems the structural representation is having vehicles that are structurally similar so it's simple er in the lower case triangle out there in the world is there a structural resemblance that would be very representational whereas if they were um cognitively imagining something different than a triangle that would have less resemblance but that gets challenging with adaptive adaptive because we're modeling ourself modeling each other blah blah blah so here's them modeling the conversation and that's where there's sort of like very complex dynamics which is how can we have a structural similarity about a conversation with another person but um it's simpler in the adaptive mirror case the content related representations encode environmental contingencies or sensory motor contingencies so like in this case the triangle it is the case that it can rotate clockwise and so this entity does have a good content related representation function if it can actually turn the dial to that way um and again that's a little bit more complex in the reciprocal case but in the unidirectional case um this red arrow is a content related representation of the actual motion in the world and then this last case is the functional representation and here it still is um

it could be more structural so like more about the actual structure of the cognitive model or it could be more about the encoding of the contingencies but the key piece is this isolation which isn't a markovian isolation this is actually the true vicarious separation of so here the person or the entity is able to engage in either structural pondering of the triangle or functional pondering of the triangle in the absence of seeing the triangle or grasping the triangle and again though that's a little bit um complex when you have multiple counterfactual cognizers asking each other questions i just added a counterfactual to your to your

to your great yeah exactly the red one is um then there's the blue one it could go a different way and then this is this is how we know that it's a true

cognizer and a true functional representation yes great call thanks a lot blue because it has to do with the vicarious nature

with not directly seeing it but it also has to do with that representation whether more structural triangular or more functional red arrow could be different so there's them having a counterfactual action contingency and here they're also imagining a counterfactual structural position

okay i'm going to go to a question from steven in the chat steven asked how do we resolve the use of the term functional as used in correlational dynamics so that's referring to some of our earlier discussions on like functional and effective connectivity in neuroimaging and the term functional in use

uh and the term functional in terms of having cognitive counterfactuals so how is the functional here related to effective and functional connectivity in neuroimaging and time series statistics

i'll give a first thought it's not that it can't be linked or connected but they are different name spaces and they're totally different the functional connectivity in the time series is um about how changes in one variable through time influence another variable through time or are associated with changes in the other one through time so it's not a mechanistic causal relationship per se but it's just that changes in one variable have an edge reflecting how changes in some other variable change that does not

engage either of the pieces that we're highlighting here which is the vicarious or standing for nature nature nor the counterfactual nature so it is a different use of the term functional functional is an overloaded word and so yeah it's not that you couldn't have a paper that was looking at functional representation and using neuroimaging and using functional and effective connectivity of brain regions while individuals were engaged in functional representation but that would a little bit be like the star was the star of the movie it's just using a word in multiple senses in a way that may give some ambiguity to those who don't parse out the different senses extremely clearly is that fair to say or does anyone have a different thought function functional can also mean teleological like the function of something essentially so there's many uses of function and i think it's um interesting that it has come up in these different spaces blue so i do agree that like they're they're disconnected but but i am able to perhaps see a relationship in like functional connectivity and also in counterfactuals like so when things are functionally connected like one drives the other right um and then the counterfactual is like when things are not functionally connected there's no there's no like you know corresponding correlation right so so in the

explore exploration of functional connectivity you have to have all of the data points like over a time series so you have to um like you understand functional connectivity through the exploration of counterfactuals does that make sense

okay how about this functional connectivity the term it is a functional representation that cognitive scientists use because it could be otherwise they're engaging and they're saying region a and b have functional connectivity because i did the time series statistics in spm and we got this value and so

that functional connectivity value is able to be used in subsequent internal cognitive representations it stands for something and it can be engaged with in a counterfactual way like what if it would

have been stronger then what would our conclusion have been or what if it would have been weaker what would it have been um and then steven wrote um these definitions these are definitions and ways to name

could be where the ontology working group could have a future role yes ontology development in active inference and how will we even make the decisions of what terms to use in what senses and

the use in what senses and how will we make the decisions of what senses and how will we make the corpus that we are looking at so pretty interesting this graphic actually i hope any of the authors or

anyone who is like interested in this topic what do you think about these doodles do you agree disagree let's go to blue's question about autopoiesis what is like interesting or that about

just really where it comes in and i think we kind of brought it up earlier in our um in the sharing of

a model like so is sharing of a generative model necessary for self-assembly or even the attempt to share a generative model is that that a necessary component of autopoiesis um and at what level like

organizationally structurally content related functionally like at all the levels um i just wonder

like how these things are related and when we were talking earlier about having a conversation and

trying to make your models align with one another through information sharing um and like we saw with

um the computational boundary of itself about self-assembly and autopoiesis uh via mike levin um

just about the informational sharing between subunits leads to the formation of a larger cognitive unit

and so um i was just wondering about model relationships and forming a larger cognitive

unit like we're talking about the conversation and then we form with complete overlap of yes we get it

we grok we're there and then we form a larger cognitive unit um and is that on all aspects is

it just related to content structure organization or function or all of the above um just what anybody

thinks about that all right so and we saw this actually in matt sims earlier paper the biological

symbiosis right yes the autopoietic system is capable of producing and maintaining itself by creating its own parts so it's kind of like the ship of theseus plus crew they're able to be modifying and

reconstructing the material basis of the system

on one hand that maintains an organizational separation of like the cell and the surrounding

in terms of the realist interpretation of a markov blanket

at the same time the autopoietic process does not need to have

well it involves ingressing and outgressing material so that does blur

the boundary and structurally it doesn't need to be the case that the autopoietic process of the cell

has like a blueprint or a structural resemblance of the cell it could just be subunits that

are non-representational they cannot engage um it doesn't even make sense to ask whether the cognitive model has structural similarity with the target because it's a non-cognitive entity and similarly on the functional side

the enzymes can't engage in counterfactuals that you know synthesize the lipids and add them to the membrane so

is information a representation

i don't know i mean we're getting getting into the quantum now

so um yeah where's where's information in this it comes up in the paper

um 18 17 times based on local information information comprises the generative model

past and present information future information information gain no additional information about internal states that's the markov definition high mutual information can be

um in the context of a non-representational generalized synchrony like the pendulums that synchronize where is information in this eightfold distinction so uh the author talks about here if i can read a quote quickly if one only appeals to the organizational aspect of representation the presence of environmental or sensorimotor motor or complex or frugal model does not matter insofar as the internal variables of the model are understood to be separated from external reality by a markov blanket and the generative model is leveraged to infer the causal structure of external reality via self-evidencing however these differences matter if one considers structural aspects and the degree of resemblance between hidden variables and environmental dynamics as opposed to action or information gathering dynamics so i don't know here here the content of representations is used to draw a further distinction within the representational view between an internalist sometimes called intellectualist or encodist versus an action-oriented perspective so

probably one of the most thrilling academic debates dean

well please don't don't judge what i'm about to say next just bear with me be kind so in this representation for example where we've got this table if we were to look at those lines each one of those horizontal lines because there's three ic and each one of those vertical lines because there's four ic and we realized and we realized them as an a standard xy graph we know that there's a relationship between x and y that every one of those horizontal lines relative to y has a zero component to it and we know that every one of those vertical lines relative to x has a zero component to it so most people will look at the blackness of that relative to the white background and they'll just see something material separating but if we actually look at it graphically and we take daniel's idea of the tetrahedron every time we round an edge and change direction on that tetrahedron there is a zero element

that allows for that boundary crossing that we again don't pay as much attention to and so from a from an auto poetic

standpoint you can actually see where something like a markov blanket both separates and leaves the potential open

for that transfer or that crossing or that transition

so when we're modeling ourselves the last paragraph of the section five of the paper they wrote arriving at an fvp as synthesis so some some non-zero number and zero view depends upon which representational criterion

we are assuming when either considering fvp central constructs or considering specific cognitive phenomena through the lens of a process theory under fvp do we include the zero piece of this or not

tentatively fully or do we willfully ignore it hence in the end the debate about fvp may reveal more about us our criteria whether we want this to be a material thing and not not include the zero aspect of it and our interest in particular facets of cognition than does about the representational status of fvp so i i just want to kind of bring that again don't judge me because i could be way wrong here and i may be overfitting but i think that the autopoietic part of it

we can see both the separation and we can see the the portal in a line if we choose to include both that's kind of what i wanted to bring up

okay very interesting dean thanks

here's the unfolded tetrahedra there's a few unfold there's a few shapes of paper that can be folded into a tetrahedra one looks more like a parallelogram and one looks more like an equilateral triangle we're just kind of like folds up like the petals of a flower so these zero lines so there's the zero point of within a facet representational and non-representational that's almost like the two sides of the paper like how many triangles are here oh four how about eight how many units how many triangle units of paint do you need to paint this four no eight two sides so the zero is like the thickness of the paper and that's also when you fold it up that's like inside versus outside of the tet the other one is like the two sides of the paper and then these zero y lines are like the movements from one face to another face where it's also thin enough as to be imperceptible especially when it's laid flat but it makes all the difference to move from one side to another

how do we use moving through zero in this eightfold or other cognitive models

how do we use that to increase the efficacy or the fluidity or the accessibility of models

does it does it help us in explaining that we're not locked out of that self-organization

loop that we can actually participate in it

is that the first thing that it tells us that we're not

we're not locked out of that ability to self-generate

that was a question that wasn't that i wasn't questioning that was it was an like an honest sincere question

i think so one supportive or a complementary component would be this is sort of a map this is like a cognitive map and someone could say this side of the paper is better than the other side and this is my favorite face but it's presented as part of a larger knowledge structure such that the learner could engage in counterfactuals like what if we were on the other side of the paper or what if we were on a different face so this allows for example individuals to reflect their understanding of the lay of the land but also communicate their preferences and as the paper shows different individuals do have different outcomes in terms of their cognitive conclusions and that's

[illegible]



Dean.

Yeah, I think, Danny, if you take the words reality and model and push them out into even more extreme tales, you could see the model.

You could see people who are pushing back against the model idea and they're arguing, you're just too rigid, right?

You're just too stuck.

You could see people that are taking the non-representational sort of reality view, the variance maintained or retained view, as you could see the people that are on the model end of the continuum saying, that's just chaos.

There's nothing about that that we can actually make sense of because the cognitive overload right now is just swamping me.

Right. And so I think what these authors were trying to do is say, well, can we move back and forth instead of getting stuck on those really far out positions in terms of this way of being able to see when we have these densities?

Right. And then how do those densities reset back to some sort of invariance and then reform?

Right. So I think that's I hope that kind of speaks to what Daniel was saying, too, because.

Sure, we can we can point out the most extreme example, like at the very end of the continuum, or we can actually talk about how we're going to draw these

unidirectional or bidirectional lines and boundary cross and I think that's what they were trying to open our thinking up to, but maybe maybe I'm just being Pollyanna here because I really like the paper.

I don't know.

Interesting point.

So here's some realism is also used in another way, which is kind of like it's a school of literature or a thread of literature.

And so then I found this paper affect of realism.

Evoke realism beyond representation.

And so this is in the case of literary expression.

So that's an activity that cognitive entities are engaged in.

And we just see a lot of terms that we talk about in act imp, like agency, affect, process.

And here it's not about reality, realism, territory.

It's like realization.

And realization is an interesting word.

And I know Verveke and others use it as well because realization, it is about the model.

But when something is realized, it becomes real in the model.

But usually we wouldn't say that something is realized like, oh, I just realized that zebras have nine legs.

It's like you engaged in a counterfactual, but is it a realization if the model is updated in a way that isn't concordant with reality?

And I agree that the paper gives us a lot of nuance to talk about these areas because we can talk about how, well, the isolation of the variables doesn't change.

Or the structure of the legs and the zebra doesn't change.

Or it can still run, right?

So why not nine legs?

Is it still a zebra?

Like there's a lot of open threads, but the way that they use model and reality in light of a lot of the other discussions we've had is really illustrative.

Blue?

Blue?

So one of my favorite things to think about in terms of the realism versus the instrumentalism is the Markov blanket.

I was looking for the picture of the person sleeping under the Markov blanket.

Like really as a partition for a system, do systems have Markov blankets?

Or can they be modeled with Markov blankets?

Like, and I always just like think of that cute little picture with the little person sleeping under the Markov blanket.

Like, do I have a Markov blanket wrapped around me?

Because that's really where it tends to break down in my mind.

Like, yes, there's a boundary,

but I'm not sure that it's a Fristin blanket

or a pearl blanket or a Markov blanket.

Like maybe it's just skin.

Yes, Dean?

I hate answering a question with another question, but in the conclusion of the paper,

the authors speak specifically to the free energy principle  
can be very heuristic.

And so I'm wondering in the instrumentalist versus realist debate,  
whether it has to necessarily get up to that abstract level  
of Markov blanket or whether we can just say that sandwiched  
between those two are rules that we demarcate,  
that we lay down and that we have act as a parsing mechanism.

I don't know. What do you think, Blue?

Like, are rules things that we can just get our hands on right today  
instead of having to go all the way up to figuring out  
what a Markov blanket is under this set of circumstances?

No, I think definitely rules, right?

So, I mean, we all agree that there are laws in physics.

Like if you drop an apple, it will fall to the floor.

It's like we all abide by these rules.

They might not apply at every scale,

but I definitely think that a rule-based construction  
or understanding,

like does that then bridge the realism versus instrumentalism debate perhaps?

So, building on the point that the ourselves in the title  
is the people reading the paper.

So, not just like people in the psych department,  
but it's the ourselves,  
the people who are in that epistemic commons.

So, using that,

but also the content-related representations  
in terms of correctness or truth.

So, if the ourselves are those who are pondering FEP,  
then the heuristic that they refer to,  
we conclude by highlighting the heuristic power of the FEP  
to advance our understanding of the notion of internal representation.

It is pretty meta

because we're representing the debate about people,  
representing the debate about representation.

However, the fact that it has a heuristic utility  
within the content-related aspect,  
at the very least,

it does have representational impact.

I'm not sure where that takes us.  
There's probably other ways to take it,  
but I think something about  
the diversity of conclusions that people have  
and how that shines a light back on us  
or is a mirror  
and how the FEP  
has been applied here  
in this paper  
as a heuristic for sense-making  
in this meta-debate  
says something, I hope.

Danielle?

I hope I'm addressing what you're saying  
and not taking this in another direction,  
but stepping back,  
I'm thinking of FEP  
as just one of many different tools of thought  
that we have  
that allow us to align our representations.  
And if it is an effective tool,  
we all start having a conversation  
that makes sense to each other,  
that's mutually intelligible.  
in this case,  
it is kind of meta  
because it's about how the academics,  
the intellectual conversation  
about representations  
have failed to converge on something.  
There is this debate.  
There are many layers of the debate.  
And so what it reveals  
is that we haven't yet converged.  
The community hasn't converged.  
And so FEP  
is this thing that we can use  
that reveals those things

and allows us to then better,  
because it's revealing these things,  
we can all attend to those things  
that were implicit  
and are now made explicit.  
We're converging our representations.  
So I think that kind of addresses  
what you're saying.  
It's about representations,  
but it's also this tool.  
Like there are many other types of tools  
that allow us to realize  
what we've been taking for granted  
and align our representations.

Thanks a lot.

Dean?

Yeah, just to add on to that,  
if you're like me,  
you think rules and tools  
are interchangeable.

And the other thing you think  
is that if it's a heuristic,  
it speaks kind of to that  
vicarious function  
in terms of content.

So I just want to throw that on top  
because I think there's  
some agreement around that.

We use the rules  
to make that next  
vicarious play  
before we make the next move.

So, I mean,  
anybody that doesn't  
is kind of kidding themselves,  
I think,  
but maybe I'm wrong.

Okay.

Here's a little bit of FEP  
in a heuristic case.  
So we've talked about  
the Helmholtz decomposition  
of vector fields,  
like in stream number 32,  
about how there's this pragmatic  
or sort of hill climbing  
or gradient descending component,  
whether you take the optimistic,  
biology,  
climbing, mountain, probable,  
go straight up the mountain of fitness,  
or the physicist's gradient descent  
into the bottom of the well.  
In either case,  
there's that irrotational component  
that's associated with pragmatism.  
And then there's this epistemic component  
that's associated  
with the solenoidal isocontour.  
So we've talked about  
that Helmholtz decomposition.  
So here's one way  
in which the FEP  
is being heuristically applied.  
So Danielle said,  
the academic conversation  
on representation  
has not converged.  
And it has not converged.  
And it has also not even cohered.  
Like there isn't a shared ontology.  
There's not a shared narrative.  
There's not shared words  
or regime of attention  
or what people are even focused on.  
And so what the FEP is doing,

literally like they did here  
with this distinction,  
it does a few things.  
It holds space for not overfitting.  
Like let's just say in 2055,  
all the people in the world  
or all the cognitive scientists,  
they all agreed.  
Oh great, we converged.  
Let's lock that in.  
But FEP even then  
will remind us like not to overfit  
and also not to center  
one side of the tetrahedra.  
We could say,  
this is the side  
that has these attributes  
or we focus on this side  
for these reasons.  
And no one goes there  
to this other side.  
Or there's a reason  
why we don't go there.  
Or we take a journey,  
we travel through there,  
but we don't live there  
or something like that.  
It brings a level of coherence  
to the discussion  
that doesn't necessarily  
have the goal  
of perfect static convergence,  
but still there'll be solenoidal flow  
and differences in opinion  
even then.  
Who said the FEP wasn't useful?  
Well, and maybe this is obvious,  
but is that not

what any good theory does?

Like Blue said,

well, gravity is true

because if I drop an apple,

it will fall.

Yeah, so the theory of gravity

does the same thing.

It allows us to converge

our representations.

It's just that the theory of gravity

is not itself about representation.

Yes.

This is like the Hall of Mirrors.

This paper and conversation

because it is like,

it's the group of people

in the Hall of Mirrors

and they're all wearing mirror suits

in a weird way.

But it is our gym

for building the kind of frameworks

that then can reduce

our uncertainty a ton

potentially in similar cases.

And the strange loop

of reflexive cognitive entities

is going to be this open-ended

about for a long time to come.

But just like how we started here

with a really complex reciprocal case

of two adaptive entities,

but then it was a lot less ambiguous here.

And I wonder if

ActInf informed system design

will frame an increasing amount of pieces,

especially in a digital world,

into this bottom tier,

offloading some of the repetitive tasks,

attention-consuming tasks,  
anxiety-producing stimuli  
into these type of interactions,  
which opens up new spaces  
for these kinds of interactions  
with the two adaptive entities.  
To return to that  
having or possessing a Markov blanket,  
just one quick point was,  
which we already even read the quote,  
which was a core aspect of FEP,  
such as the possession  
of a Markov blanket.  
So is the Markov blanket the system?  
Or does it have one?  
Those are both realist.  
Or is it modeled as having?  
Is it modeled as being?  
Those are instrumentalist ways  
of similar framings.  
And how do we know  
what the core aspects are?  
What does that stem from?  
So there's a ton to learn  
and discuss there.  
And then let's look at any,  
in just our last few minutes,  
any of the other topics.  
And if anybody wants to highlight  
one of these,  
or also maybe as we jump off  
from 36.2 to 37 and beyond,  
when we discuss free energy,  
a user's guide,  
but also we just continue  
on our paths.  
Like, why does it matter?  
Not all these papers

maybe make the argument  
for why this whole representation  
debate matters.

So what is the pragmatics  
that make this meaningful,  
make it important to fund people  
working on it,  
or the way that it influences  
the technologies that we're developing  
rather than just a pure  
info gain reduction  
of uncertainty question?

So any topics here  
or any like thoughts  
on how this will  
or could or should  
influence our action selection  
going forward?

I'm comfortable with long silences,  
but I'm going to,  
I'm going to throw this out there.

I think,

I think people will be able  
to use this tool  
to become more creative,  
not just be able  
to reduce their minimum,  
reduce or minimize  
their prediction errors,  
but actually have the confidences  
to come up with new  
and different ways  
of looking at problems  
that we probably didn't have  
in our,  
in our toolkit before  
in terms of being able  
to come up with

something different.

different.

And so I think

the first thing that it does

is it allows for a boundary

crossing that,

that maybe we constructed

in our minds before,

but we're seeing less and less

of it as being a,

a hard stop.

And there's now

maybe a little bit more,

more say in whether to go

or no,

or don't go.

So that's what I would think

that this kind of work

is,

is,

is maybe opening up

unless somebody comes

and shuts it down.

And I can't speak

to what outside forces

would find terribly threatening,

but I'm sure

there are some out there.

Outside.

Blue.

Blue.

So I'm not sure

if this has any

pragmatic value,

but I'm,

I'm certainly interested

in kind of really exploring

like sensory motor engagement

and structural representation

and maybe like synesthesia.

I know Daniel's been to Meow Wolf,

but like,

you know,

when you're playing

these invisible,

like the invisible harp,

you know,

and it's like,

could you structure a room,

like draw a map in a room

and your goal is to like,

kind of like an escape room,

but your goal is to find,

navigate your way

through based on sound,

right?

Like you ting the things

and you have to find

the right sound.

You're playing Mary Had a Little Lamb

and then you get to your prize

at the end

or you get to get out

of the room.

And so,

so I wonder creatively,

like,

like Dean was saying,

like really these maps,

these representations,

can we get literally

out of our own representation

and into the feeling

of an alternate representation?

And what would that feel like

or be like?

And that's also why  
I brought up like blindness.

Like,  
you know,  
if you ever wake up  
and like a totally,  
like my power went out  
the other night  
and it was like black.

I mean,  
no moon,  
nothing.

Like,  
it's just like,  
whoa,  
total darkness,  
right?

Like no little blue light  
from the plug  
or nothing.

Right?

So,  
in these like total darkness  
situations,  
you have this representation  
of like,  
what does my house look like?

Where's my flashlight?

Like,  
you have to be able  
to navigate yourself  
in that totally dark situation.

It's,  
which is very foreign  
to go find a flashlight.

And so I like these  
kind of boundary pushing  
experiences

and because they really  
force us to examine  
our own model  
and perhaps allow us  
to explore the kind of models  
that other people use.  
Thinking through other minds  
is this case  
with adaptive agents  
in dialogue.  
dialogue  
and it's,  
it's like thinking  
through other representations  
and this paper  
and discussion  
opens up  
not to then  
simply converge  
but opens up  
and holds  
ways  
to think  
and act  
through other  
types of representations,  
not just the same type  
but in a different  
cognitive entity.  
it comes up  
a lot  
and it'd be awesome  
to see how  
just as so many  
other themes  
that we've  
had raised  
by very prescient papers

like  
14,  
math is not the territory,  
20,  
Emperor's New Markov Blanket  
and some of the further  
nuancing  
about  
the Markovian assumptions  
this  
representational  
and non-representational  
and these words  
like organization,  
structure,  
content,  
and function  
they will come up  
again  
and maybe  
they'll be  
citing  
this paper  
maybe they  
won't  
but like  
it's almost like  
when we see  
in a future paper  
just one facet  
of the TET  
and they'll say  
here's how it is  
it's this one side  
and it's like  
well no  
there's other sides  
and we could be

inside  
or we could be  
outside  
and someone  
could still say  
I prefer this one  
but  
we know  
there's a bigger  
system  
that we can  
at least  
come to the  
table around  
Dean?  
I'm not  
I'm not  
marketing  
but it's going  
to come up  
in 37  
it's going  
to come up  
a bunch  
of times  
in 37  
so that's  
just the  
next paper  
so anyway  
hold on  
to your  
socks  
because if you  
think you have  
to wait  
for it to  
come up

again  
it's going  
to come  
up again  
next week  
don't touch  
that dial  
thanks a lot  
for the  
awesome  
sequence  
this was a  
somewhat  
complex  
or  
intellectual  
paper  
I mean  
they're all  
written  
but this one  
was like  
challenging  
topics  
a lot  
of  
different  
perspectives  
and then  
not only  
did they  
hold  
it open  
for all  
of the  
different  
perspectives  
but then

rather than  
pick a  
winner  
they flipped  
the table  
in the  
ends  
and  
posed it  
not as  
an answered  
competition  
for us  
to then  
have  
like  
domain  
specific  
clarity  
of action  
within  
but really  
a challenge  
on multiple  
levels  
for  
researchers  
who are  
familiar  
with  
FEP  
and not  
the big  
question we  
asked was  
can the  
free energy  
principle

help us  
advance  
or even  
resolve  
the long  
lasting  
debate  
on internal  
representation  
in philosophy  
of mind  
we certainly  
asked  
so if  
anybody has  
any last  
comments  
they want  
to make  
on  
anything  
they're welcome  
to  
the last  
word  
so i'll get  
this in  
real quick  
i think  
that  
if  
if  
we're still  
arguing  
about  
whether  
or not  
representation

and modeling  
is heavily  
dependent  
like that  
there is a  
dependency  
i think  
that  
the more  
philosophical  
views  
that  
we  
see  
come up  
around  
FEP  
and active  
inference  
the harder  
and harder  
it is  
to make  
the case  
that we  
can  
remove  
that  
dependency  
or  
somehow  
just  
see the  
scale-free  
aspects  
of this  
tool  
that there's

actually a  
heavy  
dependency  
on what  
we're  
going to  
get  
out  
of  
representation  
or  
non-representation  
based on  
our awareness  
or our  
realization  
that  
every time  
we cross  
a boundary  
that is  
just an  
independent  
act  
that's  
a highly  
dependent  
act  
so that  
will be  
my  
last  
word  
so  
i just  
want to  
give a  
final

comment  
you know  
we talk  
about  
instrumentalism  
and  
realism  
and then  
there's  
like  
third  
axis  
right  
of  
utility  
and  
i really  
think  
that  
this  
paper  
was  
super  
useful  
in  
prompting  
new  
questions  
in  
holding  
up  
these  
different  
views  
of  
representation  
and  
non-representation

and  
in  
these  
different  
categories  
and  
allowing  
us  
to  
really  
kind  
of  
view  
them  
through  
like  
a  
partitioned  
framework  
that  
i think  
will  
drive  
future  
questions  
and  
really  
be  
useful  
in  
helping  
researchers  
formulate  
their  
theories  
great  
danielle  
you're

still  
here  
thanks  
for  
joining  
for  
your  
first  
stream  
do

have  
any  
last  
comments  
no  
this  
was  
a  
wonderful  
introduction  
i'm curious  
how representative  
it is  
of other  
conversations  
it was  
super  
heady  
in a  
really  
pleasant  
way  
it was  
not  
representative  
or  
representational

of other  
conversations  
no  
thanks a  
lot for  
the  
awesome  
convo  
everyone's  
always  
welcome  
to  
join  
these  
discussions  
or other  
lab  
activities  
so  
thanks again  
to the  
authors  
and all  
the  
participants  
here  
see